DIGITAL FORENSICS INVESTIGATION REPORT

Network Forensics— UNODC PCAP Analysis

Case Name	PacketMaze -- UNODC Network Analysis
Case Number	2026-009
Examiner	Tej Chaute
Date	06/04/2026
Platform	CyberDefenders
Evidence File	UNODC-GPC-001-003-JohnDoe-NetworkCapture-2021-04-29.pcap, SHA256: 3cc2061959afb116aeedce2736809f28236b96e20b89b4199194f4a30a0802ba
Classification	Training

1. Case Summary

A network traffic capture was analyzed following suspicious user activity within an internal environment. The objective of the investigation was to identify communication patterns, credential exposure, file transfer activities, and external connections that could indicate potential misuse or data exfiltration.

Analysis of the PCAP file revealed that the internal host **192.168.1.26** actively communicated with multiple services including FTP, DNS, HTTP, and TLS-enabled endpoints. The user authenticated to an FTP server using cleartext credentials, browsed directories, and uploaded files. Additionally, the host initiated encrypted communications with external services, including ProtonMail, and accessed external web resources.

The observed activity demonstrates a sequence of user-driven actions involving authentication, file transfer, and external communication, with potential indicators of data handling and privacy-focused service usage.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	UNODC-GPC-001-003-JohnDoe-NetworkCapture-2021-04-29.pcap	3cc2061959afb116aeedce2736809f28236b96e20b89b4199194f4a30a0802ba	36.7 MB

3. Tools Used

Tool	Version	Purpose
Wireshark	4.6.4	Packet inspection, protocol analysis, stream reconstruction
MAC Lookup Tool	--	Vendor identification from MAC address (https://dnschecker.org/mac-lookup.php)
EXIF Tool	--	Metadata extraction from transferred files (https://exifinfo.org/)

4. Key Findings

Each finding below is supported by network artifacts extracted from the PCAP file. Timestamps reflect packet capture time in UTC.

4.1 Network Context

Field	Value
Internal Host	192.168.1.26
FTP Server	192.168.1.20
External IP	104.21.89.171
External Service	dfir.science
Encrypted Service	protonmail.com

FTP Mode	Passive (PASV)
MAC Vendor	PCS Systemtechnik GmbH
Vendor Country	United States

4.2 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	FTP Authentication	USER kali / PASS AfricaCTF2021	2021-04-30 01:01:26	Cleartext FTP credentials confirm successful authentication to the FTP server, exposing sensitive login data over the network.
2	FTP Directory Listing	/home/kali	2021-04-30 01:01:26+	The user enumerated directories on the FTP server, indicating interactive session activity and access to file system contents.
3	FTP File Upload	STOR 20210429_152157.jpg	2021-04-30 01:01:56	The user uploaded a file to the FTP server, confirming outbound file transfer activity and potential data placement on the server.
4	EXIF Metadata	Uploaded JPG File: 20210429_152157.jpg (extracted via exifinfo.org)	2021-04-30 01:01:56+	EXIF metadata extracted from the uploaded JPG identifies the originating device as LM-Q725K — provides device-level attribution for the file transfer
5	DNS Query	www.7-zip.org	2021-04-30 01:02:57	DNS resolution of www.7-zip.org indicates browsing or software lookup activity from the internal host — consistent with archiving tool research
6	TLS Handshake	SNI: protonmail.com	2021-04-30 01:04:29	The client initiated an encrypted session with ProtonMail, indicating use of privacy-focused communication services.
7	TLS Client Random	Extracted from Client Hello	2021-04-30 01:04:29	The TLS client random value uniquely identifies this specific TLS session. If the server's private key were obtained, this value — combined with the

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
				server random — would allow the full session to be decrypted using tools such as Wireshark's TLS decryption feature. Documents the session for potential future decryption if keys become available
8	HTTP Request	GET / (Host: dfir.science)	2021-04-30 01:06:39	The user accessed the dfir.science website, confirming external web browsing activity.
9	HTTP Redirect	Location: https://dfir.science/	2021-04-30 01:06:39	Server response indicates redirection to HTTPS, confirming successful connection and intended destination.
10	MAC Address	FTP server MAC: 08:00:27:a6:1f:86 (dnschecker.org/mac-lookup.php)	N/A	MAC address of FTP server resolves to vendor PCS Systemtechnik GmbH (US) — infrastructure attribution for the internal FTP server; vendor is associated with VirtualBox virtual machines
11	FTP File Retrieval	RETR accountNum.zip	2021-04-30 ~	ZIP archive named accountNum.zip retrieved from the FTP server — filename suggests financial or account-related data; retrieval over cleartext FTP represents a potential data transfer of sensitive content
12	File System Artifact	Non-standard folder: ftp (created 2021-04-20)	2021-04-20 17:53	A directory named 'ftp' was present on the server with a creation date ten days prior to the main investigation timeline — indicates earlier FTP server activity or pre-configured staging directory

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times in UTC.

Timestamp (UTC)	Artifact Source	Event Description
2021-04-20 17:53	FTP Directory Listing	Directory named 'ftp' present on FTP server — creation date ten days prior to main timeline indicates earlier server activity or pre-configured staging
2021-04-30 01:01:26	FTP Traffic	Internal host (192.168.1.26) authenticated to FTP server (192.168.1.20) using cleartext credentials (USER kali / PASS AfricaCTF2021)
2021-04-30 01:01:26+	FTP Traffic	Directory listing executed on /home/kali — user enumerated available files and folders on the FTP server
2021-04-30 01:01:56	FTP Traffic	JPG file (20210429_152157.jpg) uploaded to FTP server — EXIF metadata identifies originating device as LM-Q725K
2021-04-30 01:02:57	DNS Traffic	The system queried the domain www.7-zip.org, indicating browsing or software lookup behavior.
2021-04-30 01:04:29	TLS Traffic	A TLS handshake was initiated with protonmail.com, establishing an encrypted communication channel.
2021-04-30 01:06:39	HTTP Traffic	The user accessed dfir.science via HTTP and was redirected to HTTPS, confirming web activity.
2021-04-30 ~	FTP Traffic	ZIP archive accountNum.zip retrieved from FTP server — filename suggests financial or account-related content transferred over cleartext FTP

6. Conclusion

The forensic investigation identified a sequence of user-driven network activities originating from internal host 192.168.1.26, involving FTP authentication, file transfer, DNS queries, and encrypted external communications.

The user authenticated to the FTP server (192.168.1.20) using cleartext credentials (USER kali / PASS AfricaCTF2021), exposing sensitive authentication data to any observer on the network. Following login, directory enumeration and file operations were performed — a JPG file was uploaded whose EXIF metadata identifies the originating device as an LG LM-Q725K, and a ZIP archive named accountNum.zip was retrieved from the server. The filename of the retrieved archive suggests financial or account-related content and, combined with cleartext transmission, represents a notable data handling concern.

Subsequent activity includes a DNS query for www.7-zip.org, consistent with archiving tool usage, followed by a TLS session with protonmail.com — a privacy-focused encrypted email service. The TLS client random was extracted and documented to support potential future session decryption if the server private key were to become available. External web access to dfir.science was also confirmed.

While no explicit malicious payload or exploitation was identified, the combination of cleartext credential transmission, retrieval of a file with a sensitive-sounding name, a DNS query consistent with archive tool usage, and encrypted communication with a privacy service presents a pattern that warrants further host-based investigation to determine the nature and intent of these activities.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path. Notes were partially recorded during live analysis and completed after challenge submission.

Key methodological observations from this case:

- FTP transmits all traffic — including credentials, commands, and file contents — in plaintext. This means anyone with access to the network capture can read usernames, passwords, and transferred data directly from the stream without any decryption. SFTP or FTPS should always be used in place of plain FTP for any environment handling sensitive data.
- TLS encryption limits payload visibility but does not eliminate all forensic value. The SNI field in the Client Hello packet leaks the destination hostname even over encrypted connections. The client random value, combined with the server random, uniquely identifies a TLS session and can be used to decrypt the full session if the server's private key is later obtained — making it worth documenting even when decryption is not immediately possible.
- Passive FTP mode (PASV) creates a separate data channel on a negotiated port, distinct from the control channel on port 21. When filtering FTP traffic in Wireshark, both the control stream and the data stream must be followed to recover file transfer content — filtering on port 21 alone will miss the actual file data.
- MAC address vendor lookup identified the FTP server as a VirtualBox virtual machine (PCS Systemtechnik GmbH). In a real investigation this would be relevant context — virtual machines are commonly used as staging servers or internal lab infrastructure, and the vendor attribution helps characterise the environment.
- EXIF metadata extracted from the uploaded JPG identified the originating device as an LG LM-Q725K smartphone. In a real investigation this provides device-level attribution linking the transferred file to a specific piece of hardware, which may be significant depending on whether that device was authorised to handle the data in question.

- The filename accountNum.zip carries inherent investigative significance beyond its technical properties. Naming alone does not confirm the contents, but in a DFIR context it is sufficient to escalate scrutiny — a follow-up investigation would attempt to recover and examine the file contents from either endpoint.

If this were a real-world investigation, recommended actions would include: disabling or replacing FTP with SFTP to eliminate cleartext credential and data exposure; rotating the exposed FTP credentials immediately; conducting host-based forensics on 192.168.1.26 to determine what files were sent and received and whether accountNum.zip contained sensitive data; reviewing the ProtonMail session for context on what communications were sent; and examining the FTP server for any additional files or access logs.